

Supplemental Appendix to
*Attention and the Economic Response to Immigration
Enforcement*

Greg C. Wright
University of California, Merced

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This Supplemental Appendix collects the additional tables, figures, and diagnostics referenced in the main paper. It is provided for review and is not part of the typeset article. Sections and exhibits are numbered with an “S” prefix. Acronyms and variable definitions follow the main paper. References to “the main paper” point to *Attention and the Economic Response to Immigration Enforcement*.

S1 Event inventory and summary statistics

Table [S1](#) reports event-level summary statistics by enforcement wave. The compiled single-worksite inventory contains 55 ICE-led events satisfying the event-date, geography, and geocoding screens described in the main paper. Table [S3](#) lists the 51 events in the main analysis sample and the four single-worksite events that lack sufficient catchment POI coverage to fit the per-event distributed-lag DiD. Inclusion in the main sample requires at least 10 high-FB and 10 low-FB CBGs with sufficient POI coverage in the catchment. The four catchment-coverage exclusions are data-availability-driven rather than attention-correlated.

Table S1: Event-level summary statistics, by enforcement wave

	Full	2025–2026	2018–2019
Arrests reported at worksite	70 (128)	46 (89)	215 (218)
Spanish-ICE news spike	0.0298 (0.0242)	0.0337 (0.0235)	0.0054 (0.0100)
English-GDELT news spike	0.0146 (0.0145)	0.0163 (0.0148)	0.0036 (0.0027)
Late-window estimate (log pts)	-0.008 (0.047)	-0.009 (0.047)	-0.003 (0.046)
High-FB CBGs in catchment	118 (167)	130 (175)	43 (69)
Low-FB CBGs in catchment	134 (165)	143 (172)	84 (100)
Events	51	44	7

Notes: Cells report the cross-event mean with the standard deviation in parentheses, for the full 51-event sample and separately by enforcement wave. News spikes are the post-event peak above the pre-event baseline in the share of stories mentioning the term (see the main paper). The late-window estimate is the per-event weeks-4–8 differential in log foot traffic. High-FB and low-FB CBGs are the top- and bottom-decile Latin-American-foreign-born block groups in each event’s catchment. Arrests are missing for two events.

Table S2: Sample-flow from full event inventory to estimation subsamples

Definition	Events
Single-worksite ICE-led events satisfying inventory screens	55
– Single-worksite ICE-led but no estimable late-window estimate (insufficient catchment POI coverage) ^a	–4
Main analysis sample (single-worksite ICE-led with estimable late-window estimate)	51
<i>Estimation subsamples:</i>	
Events with reported arrest counts (severity test)	49
Events in 2024–2026 subsample (Google Trends national)	44
Events with DMA-level Google Trends coverage	41
Events with at least one in-state Mediacloud Spanish-language source	31
Events with estimable per-event civic estimate	11

Notes: “Events” is the unit. The 51-event panel is the main analysis sample for the per-event distributed-lag DiD and the headline cross-event horse race. Smaller numbers below reflect availability of specific second-stage covariates rather than re-estimation. The Mediacloud Spanish-language national series and the English-GDELT series are defined for all 51 events. Trends, DMA Trends, and in-state Spanish sources cover smaller subsamples. The civic estimate requires sufficient civic-NAICS POI density in the high-FB CBGs of an event’s catchment.

Table S3: Single-worksite ICE-led event inventory

Date	State	Event / worksite	Arrests
Main sample ($n = 51$)			
2018-04-05	TN	Southeastern Provision	97

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(Table S3 continued)

Date	State	Event / worksite	Arrests
2018-05-09	IA	MPC Enterprises (Midwest Precast Concrete)	32
2018-06-05	OH	Corso's Flower & Garden Center (Sandusky + Castalia)	114
2018-06-19	OH	Fresh Mark	146
2018-08-28	TX	Load Trail LLC	159
2019-04-03	TX	CVE Technology Group	280
2019-08-07	MS	Mississippi poultry operation (7 plants, 5 companies)	680
2025-01-23	NJ	Newark	
2025-02-03	PA	Philadelphia	7
2025-02-12	TX	Los Fresnos	2
2025-02-18	NY	Tupper Lake Pine Mill	9
2025-02-26	MS	3 J Underground LLC	7
2025-02-26	MS	Gulf Coast Prestress Partners Ltd	18
2025-02-26	NJ	North Bergen	16
2025-02-27	PA	Jumbo Meat Market	4
2025-03-27	CA	Powder & Protective Coatings	5
2025-04-02	WA	Mt. Baker Roofing	37
2025-04-22	VT	Berkshire	8
2025-04-27	CO	Colorado Springs	100
2025-05-02	NY	Lynn-Ette & Sons Farms	14
2025-05-13	FL	Wildwood	33
2025-05-20	WA	Eagle Beverage	17
2025-05-21	AL		
2025-05-27	LA	Mirabeau Water Garden stormwater project	20
2025-05-29	FL	Perla	100
2025-05-30	CA	Los Angeles	36
2025-05-30	CA	Buono Forchetta	3
2025-06-04	TX	Brownsville	25
2025-06-05	PA	Wyoming Valley Pallets Inc.	4
2025-06-06	CA	Los Angeles	44
2025-06-10	CA	Oxnard	35
2025-06-10	NE	Glenn Valley Foods	76
2025-06-11	PA	Five10 Flats restoration jobsite	17
2025-06-17	LA	Delta Downs Racetrack	84
2025-07-02	TX	Groomer's Seafood	12
2025-07-08	NJ	Alba Wine and Spirits	20

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(Table S3 continued)

Date	State	Event / worksite	Arrests
2025-07-10	CA	Glass House Farms	361
2025-07-10	TX	Taco Ole restaurant	18
2025-07-16	PA	Super Gigante	14
2025-08-20	NJ	Edison	29
2025-08-23	CT	Optimo Car Wash	7
2025-09-02	MO	Golden Apple Buffet	12
2025-09-04	GA	HL-GA Battery Company	475
2025-09-04	NY	Nutrition Bar Confectioners	57
2025-10-09	OH	Pancho's Tacos	5
2025-10-15	CT	Hamden	7
2025-10-19	ID	La Catedral Arena	105
2025-10-30	OR	Woodburn	35
2025-10-30	LA	Barrois Welding Services shipyard	25
2025-11-04	MA	Allston Car Wash	9
2025-11-14	AL	Nori Thai and Sushi	18
<i>Excluded: insufficient catchment POI coverage ($n = 4$)</i>			
2018-08-08	NE	O'Neill NE agricultural operation (3 sites)	
2025-06-04	NM	Outlook Dairy Farms	11
2026-01-15	MN	D.R. Horton	
2026-01-15	TX	El Paso	38

Table S4: Source-type audit for the single-worksite inventory

Primary cited source type	Inventory	Main	Med./low	Low spike
Prior compilation or tracker	23	22	11	14
State or local news	15	15	9	7
Agency or public record	10	9	8	4
National news	3	3	0	1
Business or address verification	2	2	2	1
Trade/compliance tracker	2	0	2	2
Total	55	51	32	29

Notes: The table classifies the primary citation recorded in the event inventory. “Inventory events” are the 55 single-worksite ICE-led events satisfying the screens described in the main paper. “Main sample” counts the 51 events with sufficient catchment POI coverage to estimate the event-level distributed-lag DiD. “Low news spike” denotes events with an English-GDELT post-event spike at or below 0.01. The table is an audit of the public-source inventory rather than a claim that the inventory is an administrative universe of all obscure raids.

S2 National versus local attention

This section examines whether the foot-traffic response tracks national or local attention, holding the search platform and query fixed and varying only the geography of measured attention. Table S5 reports the national-versus-DMA Google Trends horse race on the 2024–2026 subsample. National Trends predicts larger declines in foot traffic and remains negative when DMA Trends is included. DMA Trends has the expected negative sign when entered alone but adds little once national attention is accounted for, and reported arrests remain null in the same exercise.

Table S5: Local vs. National Coverage of Enforcement Events

	(1) Nat.	(2) + arr.	(3) + DMA	(4) DMA	(5) DMA + arr.
National Trends spike $\text{Spike}_e^{\text{Nat}}$	−0.000525*** (0.000158)	−0.000564*** (0.000167)	−0.000621*** (0.000228)		
DMA Trends spike $\text{Spike}_e^{\text{DMA}}$			+0.000074 (0.000215)	−0.000326* (0.000174)	−0.000330* (0.000184)
$\log(1 + \text{Arrests}_e)$		−0.001275 (0.005263)			+0.000589 (0.006522)
Constant	+0.010989 (0.008648)	+0.015012 (0.017686)	+0.012604 (0.008953)	+0.006794 (0.008799)	+0.005279 (0.021078)
N	44	42	41	41	39
R^2	0.215	0.232	0.239	0.085	0.079

Notes: OLS regressions of the per-event late-window estimate on national and DMA-level Google Trends spikes for “ICE raid” and, in columns (2) and (5), $\log(1 + \text{Arrests}_e)$. Both Trends measures are post-event spikes, defined as peak search interest over $\tau \in [0, 8]$ minus the baseline average over $\tau \in [-8, -5]$. The Google Trends sample is restricted to the 2024–2026 events for which weekly search data are internally consistent. The DMA measure is missing for a small set of events whose media markets have insufficient search volume. Heteroskedasticity-robust (HC1) standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S3 Geographic and sectoral spread of the response

If direct worksite disruption is the main channel, the response should concentrate near the worksite and in the raided sector; if perceived risk is the main channel, it should extend across geography and across sectors. Table S6 estimates the event study separately by distance from the worksite (Panel A) and by sector (Panel B). The decline is not confined to the immediate worksite, appearing in the 5–15 km and 15–30 km bands and remaining negative though smaller in the 30–50 km band. It is also not concentrated in the raided sector: same-sector estimates are not negative, while other-sector POIs and both consumer- and nonconsumer-facing POIs decline. The pattern is most consistent with a broad perceived-risk response rather than a localized same-sector demand shock.

Table S6: Mechanisms

Sample	Events	Mean estimate	Coef. on Spanish-ICE spike
<i>Panel A. Geographic reach, mutually exclusive distance rings</i>			
0–5 km from worksite	12	+0.031	–0.022 (0.017)
5–15 km from worksite	29	–0.013	–0.026*** (0.007)
15–30 km from worksite	36	+0.002	–0.023* (0.012)
30–50 km from worksite	40	–0.009	–0.009 (0.006)
<i>Panel B. Same-sector and broad-sector activity</i>			
Same broad sector as raided worksite	20	+0.003	+0.012 (0.017)
Other sectors	49	–0.011	–0.022*** (0.007)
Consumer-facing POIs	51	–0.007	–0.018*** (0.006)
Nonconsumer POIs	48	–0.003	–0.029*** (0.008)

Notes. Each row reports a cross-event OLS regression of a subgroup-specific late-window estimate on the standardized Mediacloud Spanish-ICE attention spike. Standard errors are heteroskedasticity-robust and shown in parentheses. Panel A defines high-FB and low-FB CBGs once within the full 50 km event catchment and then estimates the event-study separately by distance ring. Panel B estimates the event-study separately by POI bucket. Same-sector POIs have the same broad NAICS-2 code as the raided worksite. Consumer-facing POIs are NAICS-2 codes 44, 45, 72, and 81. Stars denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

S4 Additional result diagnostics

S4.1 News measures, inference, and local attention

The cross-language and cross-source validity of the headline measure can be quantified directly. The Mediacloud Spanish-ICE event spike closely tracks the English-GDELT ICE-raid spike across the full event panel and closely tracks the US national Google Trends ICE-raid spike where Trends data are available. The English-GDELT spike also tracks Spanish-language search spikes for “redada” and “deportación” in the 2024–2026 subsample. Event-level ICE news cycles propagate consistently through English news supply, Spanish news supply, and English and Spanish search demand on essentially the same schedule.

The choice of search term within the Mediacloud Spanish corpus matters. The headline uses the noun “ICE,” which has by far the cleanest event-specific signal. Alternative phrasings such as “agentes de ICE” and “redada migratoria” give the same negative sign but weaker statistical power. “Deportación” is essentially null, consistent with the term tracking the Spanish-language political-policy news cycle rather than event-specific raid coverage.

Because the headline selects one measure from this family, a correction for multiple testing is applied across the five news-volume measures. Table S7 reports Holm and Romano-Wolf family-wise adjusted p -values. The Spanish-ICE coefficient remains significant after both corrections (Holm $p = 0.003$, Romano-Wolf $p = 0.007$), as does the English-GDELT measure, while the weaker Spanish query variants do not. The Romano-Wolf adjustment, which accounts for the strong positive correlation across the measures, is smaller than the conservative Holm bound.

Table S7: Multiple-testing correction across attention measures

Attention measure	Coef	t	p	Holm p	RW p
Spanish “ICE” (headline)	-0.0184	-3.65	<0.001	0.003	0.007
English-GDELT national	-0.0154	-3.06	0.004	0.014	0.024
Spanish “agentes de ICE”	-0.0095	-1.74	0.088	0.177	0.212
Spanish “redada migratoria”	-0.0104	-1.97	0.054	0.162	0.177
Spanish “deportación”	+0.0019	+0.35	0.726	0.726	0.735

Notes: Each row is a separate random-effects meta-regression of the per-event late-window estimate on the standardized attention measure, with Knapp–Hartung inference and inverse-variance weights (same estimator as the main paper’s meta-regression); $N = 51$. The family is the set of news-volume measures discussed in this appendix. “Holm p ” is the Holm step-down family-wise adjusted p -value, which controls the family-wise error rate under arbitrary dependence. “RW p ” is the Romano–Wolf step-down adjusted p -value from a recentered pairs bootstrap (2,000 resamples of events, fixed seed), which accounts for the strong positive correlation across measures and is therefore less conservative than Holm. The headline Spanish “ICE” coefficient remains significant after both corrections.

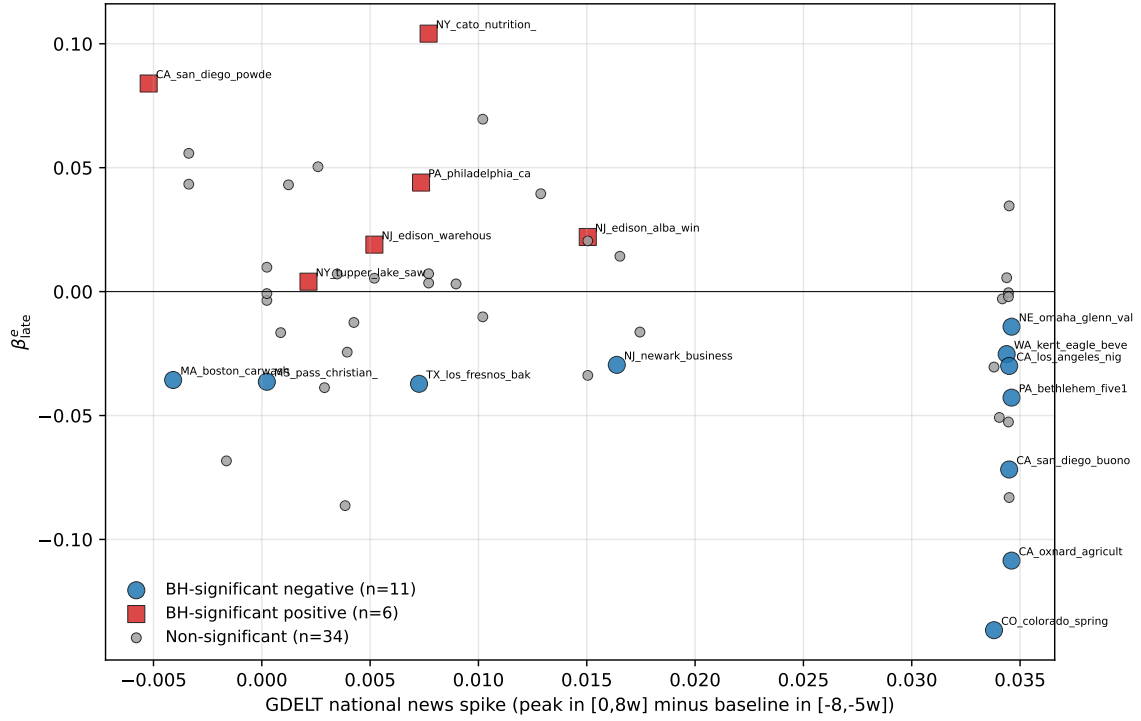
The generated-regressor concern does not drive the result. In inverse-variance-weighted WLS, where each event is weighted by the inverse of its squared standard error from the distributed-lag

DiD, the Spanish-ICE coefficient is smaller than the OLS estimate but remains statistically significant. Adding arrests leaves the result intact. An event-level bootstrap on the OLS specification also excludes zero.

The post-event attention measure also does not appear to be standing in for anticipatory media attention. For each news-volume series an immediate pre-event spike is constructed, defined as the peak over $\tau \in [-4, -1]$ minus the same $\tau \in [-8, -5]$ baseline used in the headline post-event spike. The immediate pre-event Spanish-ICE spike is uncorrelated with both the late-window estimate and the average pre-event foot-traffic coefficient. Adding it to the headline horse race leaves the post-event Spanish-ICE coefficient unchanged, and the English-GDELT measure gives the same pattern.

An event-local Spanish-news series is also constructed by restricting the Mediacloud query to outlets tagged as published in the event’s home state. The local-Spanish spike moves in the same direction as the national headline, but its predictive power is weaker and it adds little once national Spanish-language attention is included. This mirrors the same-platform national-vs.-DMA contrast within Google Trends. The predictive content of local-language local-news attention lies primarily in the part that tracks the national signal.

Figure S1: Per-event late-window estimate vs English-GDELT national news spike, by BH- $q < 0.05$ significance group



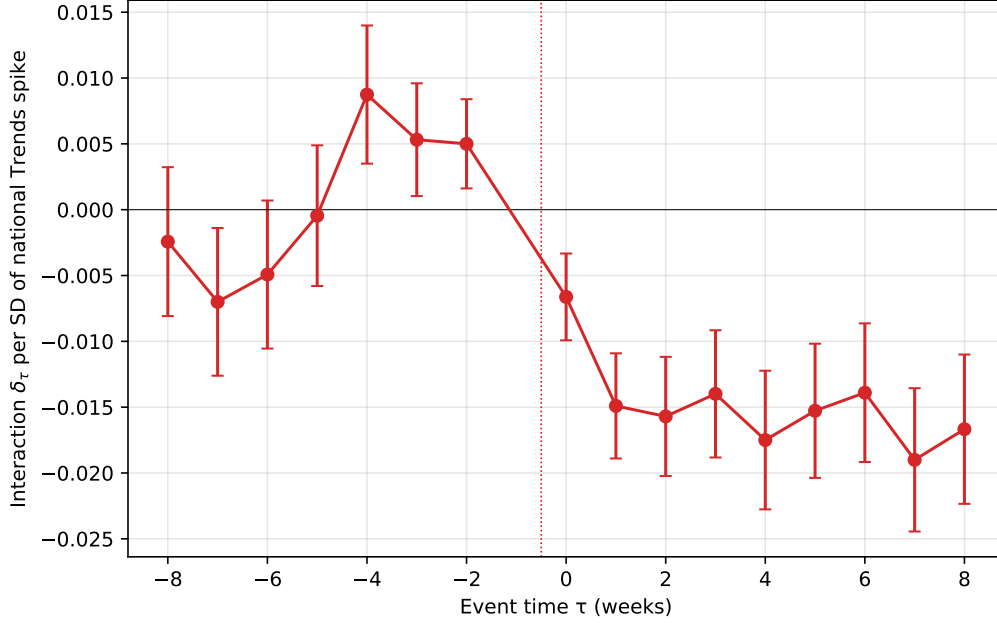
Notes. Each marker is one of the 51 fitted events. Blue circles denote events with a BH-corrected per-event significance of $q < 0.05$ and a negative late-window estimate (chilling). Red squares denote BH-significant events with a non-negative late-window estimate. Grey dots denote non-significant events. English-GDELT spikes are higher among the negative-significant group than the positive-significant group. The equivalent figure built on the headline Mediacloud Spanish-ICE measure gives the same qualitative pattern.

Table S8: Early-window robustness

	Weeks 0–2	Weeks 1–3	Weeks 4–8
<i>Panel A. Event-estimate distribution</i>			
Mean estimate	−0.001	−0.004	−0.008
Median estimate	0.000	−0.003	−0.003
Share negative	0.51	0.55	0.57
<i>Panel B. Cross-event attention gradient</i>			
Pearson r , Spanish-ICE spike	−0.263	−0.375	−0.477
OLS coef., Spanish-ICE spike	−0.323*	−0.524***	−0.919***
	(0.174)	(0.191)	(0.256)
OLS coef., Spanish-ICE spike, with arrests	−0.388**	−0.592***	−0.999***
	(0.181)	(0.202)	(0.265)
<i>Panel C. Corroborating English-news measure</i>			
Pearson r , English-GDELT spike	−0.227	−0.333	−0.416
OLS coef., English-GDELT spike	−0.466	−0.774**	−1.338***
	(0.302)	(0.333)	(0.438)
N	51	51	51

Notes. Each column summarizes the same per-event distributed-lag DiD path over a different post-event window. Week 0 is the calendar week containing the enforcement event and can include pre-event days. The weeks 1–3 window excludes the event week. The weeks 4–8 column is the headline persistent-response window used in the main horse race. OLS rows report single-regressor HC1 standard errors in parentheses, except for the “with arrests” row, which also includes $\log(1 + \text{Arrests}_e)$ as a control. The arrests-control regressions use 49 events. Stars denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure S2: Pooled event-time attention interaction



Notes. Estimates from a single pooled regression on the stacked CBG-week panel.

$\log(1 + y) = \alpha_{c,e} + \lambda_{w,e} + \sum_{\tau \neq -1} [\gamma_\tau + \delta_\tau \text{Spike}_e^z] \cdot \text{HighFB}_c^e \cdot \mathbf{1}[t - T_0^e = \tau] + \varepsilon$, where Spike_e^z is the standardized event-level national Google Trends spike. The figure plots δ_τ . Standard errors are clustered at the CBG level.

Whiskers are $\pm 1.96 \times \text{SE}$. Reference week is $\tau = -1$. $N = 204,510$ CBG-week observations. The pre-event coefficients are not perfectly flat, so the figure is treated as descriptive rather than as an identifying test.

Because the pre-event coefficients in Figure S2 are not flat, this pooled time path is treated as descriptive, and its sensitivity to violations of parallel trends is quantified. Table S9 applies the relative-magnitudes sensitivity analysis of Rambachan and Roth [2023] to the late-window average of the pooled event study. The robust confidence set for the interaction δ_τ already includes zero once the allowed post-event deviation from parallel trends reaches roughly one-tenth of the largest pre-event deviation (a breakdown $\bar{M} \approx 0.1$). The pooled time path therefore does not support identification on its own. Identification instead rests on the main paper’s cross-event meta-regression, which the design-validity checks below show is not contaminated by differential pre-trends, and on its instrumental-variables estimates.

Table S9: HonestDiD pre-trend sensitivity of the pooled event study

Robust 95% CI (late-window avg)	
<i>Panel A: Interaction δ_τ (attention gradient, Figure S2), est. = -0.0165</i>	
Parallel trends ($\bar{M} = 0$)	[-0.0212, -0.0118]
$\bar{M} = 0.1$	[-0.0277, -0.0057]
$\bar{M} = 0.2$	[-0.0356, 0.0021]
$\bar{M} = 0.3$	[-0.0441, 0.0103]
$\bar{M} = 0.4$	[-0.0479, 0.0187]
$\bar{M} = 0.5$	[-0.0479, 0.0272]
$\bar{M} = 1.0$	[-0.0479, 0.0479]
$\bar{M} = 1.5$	[-0.0479, 0.0479]
$\bar{M} = 2.0$	[-0.0479, 0.0479]
Breakdown $\bar{M}$	0.10
<i>Panel B: Main γ_τ (average response), est. = -0.0023</i>	
Parallel trends ($\bar{M} = 0$)	[-0.0076, 0.0029]
$\bar{M} = 0.1$	[-0.0110, 0.0063]
$\bar{M} = 0.2$	[-0.0156, 0.0109]
$\bar{M} = 0.3$	[-0.0207, 0.0161]
$\bar{M} = 0.4$	[-0.0261, 0.0215]
$\bar{M} = 0.5$	[-0.0315, 0.0268]
$\bar{M} = 1.0$	[-0.0535, 0.0535]
$\bar{M} = 1.5$	[-0.0535, 0.0535]
$\bar{M} = 2.0$	[-0.0535, 0.0535]
Breakdown $\bar{M}$	0.00

Notes: Rambachan–Roth (2023) relative-magnitudes sensitivity for the pooled event study (national Google Trends measure, 7 pre-periods, 9 post-periods). The target is the late-window average ($\tau \in [4, 8]$). $\bar{M}$ bounds the largest post-treatment deviation from parallel trends by $\bar{M}$ times the largest observed pre-treatment deviation; $\bar{M} = 0$ is the standard CI under exact parallel trends. The breakdown $\bar{M}$ is the largest value at which the robust 95% confidence set still excludes zero. Confidence sets use the conditional least-favorable method.

S4.2 Clean-event robustness

A contamination metric is computed for each event. The metric is the number of other in-sample events whose dates fall within ± 8 weeks of event e and whose CBG catchments overlap with e 's. Of the 51 main-sample events, 34 have zero contaminating neighbors. Of those 34, 24 fall in the 2024–2026 subsample for which the pre-event state-Trends baseline is internally consistent (see the main paper). On this subsample the saturation-test correlation between the late-window estimate and the pre-event state-Trends baseline strengthens slightly relative to the full 2024–2026 Trends sample, in the same direction and with the same statistical strength. The leave-one-out diagnostic for the headline Spanish-ICE horse race keeps the coefficient close to the full-sample point estimate when any single event is dropped. The corresponding English-GDELT leave-one-out diagnostic

leads to the same conclusion.

Table S10 addresses the concern that the cross-event heterogeneity reflects differences in the validity of the event-study design rather than differences in the underlying response. Panel A regresses each per-event measure of design quality, namely the joint F statistic for pre-event trends, the magnitude of the pre-event and placebo coefficients, the number of block groups in the catchment, the standard error of the estimate, and the contamination metric, on the standardized attention spike. None is significantly related to attention, so high- and low-attention events do not differ systematically in how clean their comparisons are. Panel B shows that the attention coefficient in the meta-regression is unchanged when these design measures are added as controls. Panel C recomputes the heterogeneity statistics and the attention gradient on progressively cleaner subsamples. Restricting to the 34 uncontaminated events, and then to the 23 events that are also free of detectable pre-trends, leaves I^2 near 60 percent and preserves the attention gradient, though statistical power falls in the smallest subsample.

Table S10: Design validity: per-event design quality is orthogonal to attention

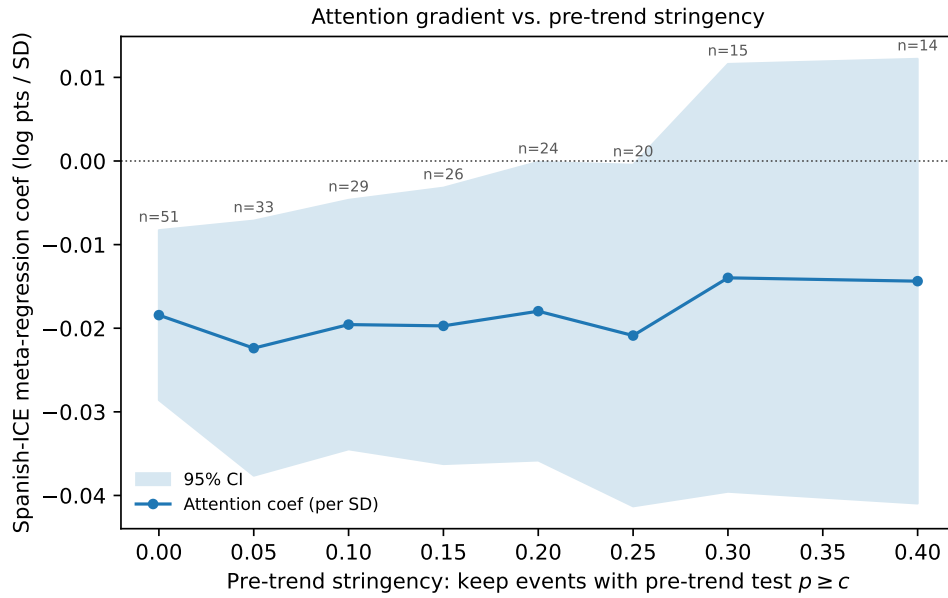
<i>Panel A: Design metric regressed on the standardized Spanish-ICE spike (HC1)</i>					
Design metric	N	slope/SD	t	p	R^2
Pre-trend joint F (non-flatness)	51	-217.6642	-1.04	0.301	0.047
pre-event coef	51	-0.0056	-1.31	0.191	0.045
placebo response	51	-0.0060	-1.25	0.210	0.035
# CBGs in catchment	51	-9.3960	-0.24	0.810	0.001
Per-event SE (imprecision)	51	-0.0005	-0.11	0.915	0.000
# contaminating events	51	0.1667	1.04	0.296	0.025
Frac. catchment shared	51	0.0256	0.51	0.611	0.004
<i>Panel B: Attention coefficient in the meta-regression, with vs. without design controls</i>					
	N	coef/SD	SE	t	p
Spanish-ICE spike (per SD), no controls	51	-0.0184	0.0051	-3.65	0.001
+ design controls	51	-0.0185	0.0054	-3.42	0.001
<i>Panel C: Heterogeneity on clean subsamples</i>					
Subsample	N	I^2	τ	spike p	R^2_{meta}
All events	51	69.7%	0.0287	0.001	30.1%
No contamination	34	61.1%	0.0309	0.027	14.8%
No contam. & flat pre-trends	23	62.8%	0.0301	0.051	16.6%

Notes: Panel A regresses each per-event design-quality metric on the standardized Spanish-ICE news spike with HC1 standard errors; a null slope means design quality does not vary systematically with attention. Panel B shows the random-effects meta-regression attention coefficient is unchanged when standardized design metrics (pre-trend F , # CBGs, # contaminating events) are added as controls. Panel C recomputes the heterogeneity statistics and the attention meta-regression on progressively cleaner subsamples (no overlapping-catchment contamination; additionally non-rejected flat pre-trends at $p > 0.10$). τ is in log points.

Figure S3 traces the attention gradient as the sample is restricted to events with progressively

flatter pre-trends. The meta-regression coefficient stays between -0.018 and -0.022 log points per standard deviation and remains negative and statistically significant as the pre-trend cutoff tightens from the full sample to the events that fail to reject flat trends at $p \geq 0.25$, a restriction that discards 60 percent of events. The confidence interval widens as the sample shrinks while the point estimate is stable. Only at the two strictest cutoffs, which retain about 14 events, does the coefficient attenuate to roughly -0.014 and lose significance, consistent with a loss of power rather than a change in the underlying relationship. The attention gradient is therefore not an artifact of the events whose pre-trends are least credible.

Figure S3: Attention gradient by pre-trend stringency



Notes. Each point is the random-effects meta-regression coefficient on the standardized Spanish-ICE attention spike, estimated on the subsample of events whose pre-event joint trend test fails to reject flat trends at $p \geq c$ (higher c requires flatter pre-trends). The spike is standardized once on the full 51-event sample, so the coefficient is in constant per-standard-deviation units. The shaded band is the 95 percent Knapp-Hartung confidence interval and labels give the number of events retained at each cutoff.

Table S11 reports the attention gradient estimated separately by enforcement wave. Within the 2025–2026 wave alone, which contains 44 of the 51 events, the gradient is at least as strong as in the full sample, so the cross-event relationship is not an artifact of the structural differences between the two enforcement eras. The 2018–2019 wave has only seven events, too few for inference, but its slope has the same sign.

Table S11: Attention gradient by enforcement wave

Sample	N	Spanish-ICE coef (per SD)	p	I^2
Full sample	51	−0.0184	0.001	69.7%
2025–2026 wave	44	−0.0199	0.001	73.1%
2018–2019 wave	7	−0.0149	n/a	n/a

Notes: Random-effects meta-regression of the per-event late-window estimate on the standardized Mediacloud Spanish-ICE spike, estimated separately by enforcement wave, with Knapp–Hartung inference. The spike is standardized within each sample. The 2018–2019 wave has only seven events, too few for inference, so its slope is reported descriptively and I^2 and p are omitted.

The gradient is also unlikely to be an artifact of mismeasured operational scale. Table S12 shows that the attention coefficient is essentially unchanged when arrests, operation type, and sector are added as controls, that attention dominates arrests in a per-standard-deviation horse race, and that the small arrest coefficient remains small relative to the attention coefficient even under generous corrections for measurement error in arrests. Arrests are the only continuous operational-scale measure the shock inventory records, so it is controlled for directly rather than by building a scale index from a single variable, with operation type and sector capturing composition.

Table S12: Operational scale and the attention gradient

	Attn. coef	(KH SE)	p	R^2_{meta}	N
<i>Panel A: Attention coefficient as scale controls are added</i>					
Spanish-ICE spike	-0.0184	(0.0051)	0.0006	30.1%	51
+ log arrests	-0.0192	(0.0053)	0.0007	25.1%	49
+ arrests, type, sector	-0.0190	(0.0074)	0.0141	15.1%	49
English-GDELT spike	-0.0154	(0.0050)	0.0036	18.5%	51
+ log arrests	-0.0161	(0.0052)	0.0035	16.9%	49
<i>Panel B: Per-SD horse race, attention vs. log arrests</i>					
Attention spike (per SD)	-0.0192	(0.0053)	0.0007	25.1%	49
Log arrests (per SD)	-0.0040	(0.0055)	0.4675	—	—
<i>Panel C: Arrests de-attenuation (single-regressor errors-in-variables)</i>					
reliability $\rho = 1.0$ (no measurement error)	-0.0030	implied arrests coef / attention coef = 0.16			
reliability $\rho = 0.7$	-0.0042	implied arrests coef / attention coef = 0.23			
reliability $\rho = 0.5$	-0.0059	implied arrests coef / attention coef = 0.32			
reliability $\rho = 0.4$	-0.0074	implied arrests coef / attention coef = 0.40			

Notes: Random-effects meta-regressions of the per-event late-window estimate on standardized regressors, with Knapp–Hartung adjusted inference and inverse-variance weights from the clustered linear-combination standard errors (same estimator as the main paper’s meta-regression). Arrests are the only continuous operational-scale measure recorded in the shock inventory: fines are unrecorded for every event, and operation end-dates are recorded for only 1 of 51 events (and reflect case resolution rather than operation length), so arrests are controlled for directly rather than building a scale index from one variable. Operation type/composition controls are worksite operation, sector agriculture, sector food, sector manufacturing, sector construction. Publicity is a visibility measure and is deliberately excluded from the scale block. Panel C reports the implied true single-regressor arrests coefficient b_{arr}/ρ under classical measurement error with reliability ρ ; even at $\rho = 0.5$ the implied coefficient is small relative to the attention coefficient. Attention coefficients are per standard deviation of the spike; the arrests specifications have $N = 49$.

The headline meta-regression is also robust to how its inference is conducted. Table S13 reports a randomization-inference p -value that permutes the attention spike across events ($p = 0.0006$), leave-one-out estimates that remain negative and significant in all 51 fits, and the heterogeneity standard deviation under three estimators (DerSimonian–Laird, REML, and Paule–Mandel), all between 0.025 and 0.031 log points. The gradient is not an artifact of the normal approximation underlying Knapp–Hartung inference, of a single influential event, or of the τ^2 estimator.

Table S13: Inference robustness for the attention meta-regression

<i>Panel A: Attention slope, alternative inference</i>	
Slope (per SD)	-0.0184
Knapp–Hartung p	0.0006
Randomization-inference p	0.0006
R^2_{meta}	30.1%
<i>Panel B: Leave-one-out and influence</i>	
LOO slope range	[-0.0202, -0.0161]
LOO fits significant at 5%	51 of 51
Most influential event (drop)	slope $\rightarrow$ -0.0161
<i>Panel C: Heterogeneity SD τ by estimator</i>	
τ (DerSimonian–Laird)	0.0254
τ (REML)	0.0287
τ (Paule–Mandel)	0.0314
I^2	69.7%

Notes: $N = 51$. Panel A compares the Knapp–Hartung p -value for the random-effects meta-regression slope (per-event late-window estimate on the standardized Spanish-ICE spike, inverse-variance weighted) with a randomization-inference p -value that permutes the spike across events 10,000 times and recomputes the slope; the permutation test makes no normality or large-sample assumption. Panel B drops each event in turn: the slope range, the number of leave-one-out fits that remain significant at 5%, and the slope when the single most influential event is dropped. Panel C reports the heterogeneity standard deviation τ under three estimators. The slope is negative and significant under permutation inference and in every leave-one-out fit, and τ is stable across estimators.

The gradient also does not reflect how events entered the inventory. Table S14 classifies each event by the type of its source citation. Events documented through news outlets carry higher attention spikes than those documented through official or other channels, but discovery source does not predict the per-event estimate, and the attention gradient is, if anything, stronger under discovery-source fixed effects.

Table S14: Discovery-source robustness

<i>Panel A: Attention spike and estimate by source type</i>			
Source type	Events	Mean spike	Mean β_{late}
news/web	32	0.0340	-0.0094
official	9	0.0349	+0.0050
other	10	0.0116	-0.0156
<i>Source $\rightarrow$ spike, joint F</i>		5.25	$p = 0.01$
<i>Source $\rightarrow \beta$, joint F</i>		1.63	$p = 0.21$
<i>Panel B: Attention gradient with source fixed effects</i>			
	Slope	(KH SE)	p
Baseline	-0.0184	(0.0051)	0.0006
+ discovery-source FE	-0.0228	(0.0051)	0.0001

Notes: $N = 51$. Each event is classified by the type of its inventory source citation (official government, news/web, advocacy/social, or other; categories smaller than 5 events are pooled into “other”). Panel A reports the mean attention spike and mean late-window estimate by source type and the joint F from regressing the standardized spike, and the estimate, on source dummies. Panel B is the random-effects attention meta-regression with and without discovery-source fixed effects. The source citation is a proxy for the discovery channel, not a measured discovery process; this is a selection probe, not an identifying control.

S5 Instrument robustness and balance

Table S15: First-stage relationship between news pressure and attention

	Mediacloud Spanish-ICE		English-GDELT	
	(1) Foreign	(2) US disaster	(3) Foreign	(4) US disaster
Foreign disaster days (std.)	-0.0168*** (0.0023)		-0.0105*** (0.0014)	
Cross-state disaster days (std.)		-0.0078*** (0.0015)		-0.0031*** (0.0009)
First-stage F	52.66	28.44	53.12	12.85
R^2	0.482	0.103	0.520	0.046
N	51	51	51	51

Notes: The dependent variable is the indicated post-event attention spike. Each column reports a single-instrument first stage with the instrument standardized. The foreign-disaster measure counts days in the ICE event’s post-event window that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The US-disaster measure counts major natural disasters in a different US state from the ICE event. Heteroskedasticity-robust standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S16: Competing-news IV robustness: foreign disasters and controls

	(1) Foreign	(2) + sev./era	(3) + event ctrls	(4) Both IVs
Attention coef.	−1.059*** (0.338)	−1.123*** (0.341)	−0.993** (0.475)	−1.052*** (0.340)
<i>t</i> -statistic	−3.13	−3.29	−2.09	−3.09
First-stage <i>F</i>	52.66	77.83	24.80	25.78
Sargan over-id (<i>p</i>)	—	—	—	0.830
Foreign-disaster IV	Yes	Yes	Yes	Yes
US-disaster IV	No	No	No	Yes
log(1 + arrests) + 2025+	No	Yes	Yes	No
Publicity + sector ctrls	No	No	Yes	No
<i>N</i>	51	49	49	51

Notes: The dependent variable is per-event β_{late}^e and the endogenous regressor is the Mediacloud Spanish-ICE attention spike. Columns (1)–(3) use the foreign-disaster instrument alone. Column (4) uses the foreign-disaster and US-disaster instruments jointly. Severity/era controls are log(1 + arrests) and an indicator for events in 2025 or later. Event controls add the event-publicity flag and broad sector indicators for agriculture/dairy, restaurant/food, manufacturing/industrial, car wash, and construction. First-stage statistics are robust Wald *F* tests for the excluded instruments. Heteroskedasticity-robust standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S16 shows that the foreign-disaster IV result is not an artifact of obvious event-level controls. Adding log(1 + arrests) and a second-Trump-era indicator leaves the estimate close to the headline foreign-disaster estimate. Adding event-publicity and broad-sector controls attenuates the estimate but preserves the sign and statistical strength. The overidentified specification with foreign disasters and out-of-state U.S. disasters also leads to the same conclusion.

Table S17: Balance check: foreign disaster news pressure and event characteristics

Covariate	<i>N</i>	Coef. on foreign disasters (cond.)	<i>p</i>
Log arrests	49	+0.296	0.034
Pre-event beta	51	−0.002	0.624
Backward placebo beta	51	+0.004	0.674
Cross-state disaster days (std.)	51	+0.486	0.003
Joint test of listed covariates	49	$F = 4.35$	0.006

Notes: Each row reports the coefficient on standardized foreign-disaster days in a regression of the listed predetermined event characteristic on foreign-disaster days, second-Trump-era, event-publicity, and broad-sector controls. The foreign-disaster measure counts days in the ICE event’s post-event window that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The joint test reverses the regression and asks whether the listed covariates jointly predict foreign-disaster days after the same era, publicity, and sector controls. Heteroskedasticity-robust inference is used. The table is a balance diagnostic, not a proof of exclusion.

Table S17 addresses the most direct timing concern. The question is whether foreign-disaster exposure is proxying for observable differences in which events occur when. The timing measure

is related to the U.S.-disaster measure and to reported arrests, which is why the specification with both instruments and controls is reported. It is not related to pre-event foot-traffic trends or the placebo response.

Table S18 pressure-tests the exclusion restriction directly. The restriction requires that foreign-disaster news pressure shift the neighborhood response only by crowding out coverage of the enforcement action. Because the outcome is a within-catchment differential, U.S.-wide disaster shocks difference out, and the main residual concern is a home-region channel in which a disaster in a country tied to the treated population affects those neighborhoods directly. The qualifying disasters are predominantly Asia-Pacific (73 percent), with the remainder elsewhere outside Latin America and the Caribbean, none with a material tie to the Mexican or Central-American-origin population. Restricting the instrument to Asia-Pacific disasters alone leaves the estimate close to the headline with a first-stage F of 47, and dropping the largest U.S. non-Latino diaspora origin countries preserves a significant estimate. The Africa-and-Europe-only slice has too few exposed events to support a first stage. The overidentified specification that adds U.S. cross-state disasters, which would violate the exclusion through a different channel, does not reject the Sargan test ($p = 0.83$). The instrument is unrelated to the pre-event coefficient and the placebo response, and it correlates with reported arrests and, mechanically, with the U.S.-disaster measure, which is why arrests are controlled for and the two disaster instruments reported jointly.

Table S18: Exclusion-restriction pressure tests for the foreign-disaster IV

	Coef.	(SE)	First-stage F	Disasters
<i>Panel A: Geographic restriction of the instrument</i>				
Baseline (ex-LatAm/Caribbean)	-1.059***	(0.338)	52.7	113
Asia-Pacific only	-1.240***	(0.402)	46.6	83
Africa / Europe / MENA only	-0.520	(0.857)	5.6	27
Drop largest US (non-Latino) diaspora origins	-0.889**	(0.439)	25.8	69
<i>Panel B: Overidentification and balance</i>				
Foreign + US disasters (over-id)	-1.052***	(0.340)	25.8	
Sargan over-id p	0.830			
Balance joint F (covariates $\rightarrow$ instrument)	$F = 4.35$, $p = 0.006$			
OLS reference	-0.919***	(0.256)		

Notes: The endogenous regressor is the Mediacloud Spanish-ICE attention spike and the outcome is per-event β_{late}^e . Panel A re-estimates the foreign-disaster IV restricting the instrument to disasters by region of the affected country. Of the 113 qualifying red-alert non-Latin-American sudden disasters, 83 (73%) are in the Asia-Pacific and the remainder elsewhere outside Latin America and the Caribbean, none with a material tie to the Mexican or Central-American-origin population. The diaspora row drops the largest US non-Latino foreign-born origin countries (the Philippines, India, China, Vietnam, Korea). Panel B reports the overidentified specification using the foreign-disaster and US-disaster instruments jointly with its Sargan test, and the joint test that predetermined event characteristics predict the instrument. First-stage statistics are robust Wald F tests. Heteroskedasticity-robust standard errors. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S19 unpacks the joint balance test. The rejection is driven almost entirely by the mechanical correlation between foreign-disaster timing and the cross-state U.S.-disaster measure, which is the second instrument rather than a confounder of foot traffic; the predetermined design-quality covariates, the pre-event coefficient and the backward placebo, are each individually unrelated to the instrument. The foreign-disaster second-stage estimate is stable when these covariates are added as controls, moving from -1.06 to -1.04 , so the balance result reflects a controllable timing correlation rather than a channel that drives the estimate.

Table S19: IV balance: what drives the joint test, and conditional-IV survival

<i>Panel A: Loadings of predetermined covariates on the instrument (per SD)</i>			
	Coef	<i>t</i>	<i>p</i>
Log arrests	+0.237	+1.26	0.208
Pre-event β	-0.129	-1.14	0.255
Backward placebo β	-0.081	-0.47	0.637
Cross-state disaster days	+0.338	+3.05	0.002
<i>Joint F (all four)</i>	4.35		0.006

<i>Panel B: Foreign-disaster IV second stage, conditioning on the imbalance covariates</i>				
	Coef	(SE)	First-stage <i>F</i>	<i>N</i>
No controls (baseline)	-1.059	(0.338)	52.7	51
+ imbalance covariates	-1.056	(0.324)	62.0	49
+ imbalance, era, publicity	-1.040	(0.359)	92.4	49

Notes: Panel A regresses the standardized foreign-disaster instrument on the four predetermined covariates that enter the appendix joint balance test (Table S17), with the same era, publicity, and broad-sector controls; coefficients are per standard deviation of each covariate. The joint F reproduces the appendix value and identifies which covariate carries the rejection. Panel B re-fits the foreign-disaster 2SLS coverage-on-foot-traffic estimate, adding the predetermined covariates as controls (the cross-state disaster measure is the second instrument, not a control). The second-stage coefficient is stable across specifications, so the balance result reflects a controllable mechanical correlation rather than a violation that drives the estimate. The conditional specifications drop two events with missing arrest counts ($N = 49$). Heteroskedasticity-robust inference throughout.

S6 Exposure-measure robustness

The headline high-FB indicator uses share of Latin-American foreign-born population (ACS B05006) at the CBG level. This is a broad measure because it counts naturalized US citizens alongside non-citizens, all Latin-American national-origin groups alongside the highest-enforcement-risk Mexican and Central American populations, and all adults alongside the US-citizen children of foreign-born parents. To test whether the chilling response is concentrated in narrower subgroups most directly exposed to deportation risk, the per-event distributed-lag DiD is refit with three sharper exposure indicators. These are share Hispanic non-citizen adults, share foreign-born from Mexico, El Salvador, Guatemala, Honduras, or Nicaragua, and share US-citizen children under 18 with at least one foreign-born parent.

Table S20: Exposure-measure robustness, per-event estimate on Spanish-ICE spike by high-FB indicator

Exposure indicator	N	mean estimate	sd estimate	r vs base	coef	t	R^2
Latin-Amer. FB (baseline, B05006)	51	-0.011	0.044	+1.000	-0.92	-3.59	0.227
Hispanic non-citizen (B05003I)	51	-0.007	0.046	+0.440	-0.17	-0.41	0.006
Mex. + Cent. Amer. FB (B05006)	51	-0.006	0.043	+0.716	-0.58	-1.98	0.089
Mixed-status kids (B05009)	49	-0.001	0.040	+0.551	-0.29	-1.05	0.027

Notes: For each exposure indicator, the per-event distributed-lag DiD (main paper) is refit, defining high-FB CBGs as the top-decile and low-FB CBGs as the bottom-decile within each event’s catchment by the indicated variable. “Mean estimate” and “sd estimate” are the cross-event distribution of the resulting late-window estimates. “ r vs base” is the cross-event correlation of the refined-indicator estimates with the baseline estimates. The last three columns report the single-regressor OLS of the late-window estimate on the Mediacloud Spanish-ICE spike with HC1 standard errors.

All three refinements produce the same negative sign as the baseline at smaller mean magnitudes and substantially weaker cross-event predictive power for the Spanish-ICE attention spike. The pattern suggests that the cross-event response operates at the broad community level rather than narrowly at the personal-deportation-risk level. This weakening is not mechanically driven by fewer POIs in the sharper top-decile cells. Top-decile POI density under the sharper indicators is within ± 6 percent of the baseline.

S7 Venue-coverage and civic-mechanism checks

Advan’s POI panel does not measurably capture two venue categories that are central to the sensitive-locations policy and to the broader immigration-fear literature. Religious organizations (NAICS-3 = 813) contain 4,025 POIs across the 26-state catchment universe, of which 3,817 (95 percent) are the Independent Community Bankers of America contamination documented in Appendix S8. After stripping obvious mislabels, only a very small number of panel POIs correspond to real religious congregations. Healthcare venues are absent. NAICS-2 = 62 has zero POIs in the panel. The civic-versus-commercial decomposition therefore captures the schools and parks/museums response but is silent on churches and hospitals.

Whether the civic-vs.-commercial differential is itself moderated by the cross-event attention spike is also tested. With the English-GDELT spike, the within-event civic-minus-commercial differential is -0.153 in the low-spike group and -0.051 in the high-spike group. With the Mediacloud Spanish-ICE spike, the pattern is similar. The differential is -0.162 in the low-spike group and -0.060 in the high-spike group. The civic-vs.-commercial gap is wider in low-attention events than in high-attention events under both measures, the opposite of what an attention-moderated mechanism would predict.

S8 POI data-quality audits

S8.1 Advan and SafeGraph POI classification

This section discusses non-trivial point-of-interest classification errors in the underlying SafeGraph POI universe. These include NAICS misclassification across sectors, POIs identified as active businesses that are in fact vacant or demolished structures, and visit-count series that are mathematically inconsistent with the POI’s physical footprint.

S8.2 Geometric POI flags

Two independent quality flags are constructed for all Advan POIs in the catchments of every event in the sample, covering approximately 1.54 million distinct POIs across the event catchments used in the audit. Building-containment is tested against Microsoft Building Footprints, and freeway-proximity is tested against TIGER Primary and Secondary Roads. A POI receives the freeway-traffic flag if its lat/lon falls outside any building footprint and is within 50 meters of an interstate centerline. A POI receives the remote flag if it falls outside any building, more than 100 meters from the nearest building, and more than 200 meters from any S1100 or S1200 road.

Across the 26-state panel, 7,881 POIs receive the freeway-traffic flag (0.5 percent) and 20,324 receive the remote flag (1.3 percent), for a combined drop rate of about 1.8 percent. The flagged set catches the expected error mode. In California, the five most-visited flagged POIs are the Port of Los Angeles, the Port of Long Beach, an Irwindale aggregates quarry, the Long Beach port-adjacent industrial complex, and the Port of Los Angeles San Pedro terminal. The flagged POIs are disproportionately in low-Latino-foreign-born CBGs rather than high-Latino-foreign-born CBGs, so removing them shrinks the comparison group rather than the treatment group. Re-estimating the per-event late-window effect on the cleaned panel shifts the cross-event mean by 0.0004 log-points. Three of 51 events flip sign, all with absolute estimates below 0.01 in either direction. The mean absolute change per event is 0.0056 log-points. The information-shock finding does not depend on the small subset of POIs that fail the building-containment and freeway-proximity checks.

S8.3 POI verification and sector recoding

The geometric check is supplemented with hand-verification of 100 randomly sampled California POIs, stratified by Veridion firm-registry match quality. Of the 100, 96 correspond to a real business or commercial property at the listed address. The four exceptions are all residential single-family homes that Advan labels with a generic NAICS-category string, and all four fall in the no-Veridion-match tier where Advan errors should be concentrated. No POI in the sample was a demolished structure or a business that never existed.

A second verification of 500 California POIs stratified by NAICS-3 was designed to test whether classification noise could reshuffle the civic-versus-commercial buckets. The Advan-bucket $\times$ true-bucket cross-tabulation shows that the directional contamination that would erase the mechanism

finding is essentially zero. Only 1 of 200 civic-labeled POIs is actually commercial. The audit did identify a consequential bug in NAICS-3 = 813. In California, 110 of the 118 Advan POIs in NAICS-3 = 813 are labeled “Independent Community Bankers of America” but correspond to individual community-bank branch addresses. A further three NAICS-813 records are sports stadiums. NAICS-3 = 813 is therefore dropped from the civic bin in the civic-versus-commercial decomposition. Including 813 attenuates the civic estimate toward zero by mixing in community-bank visitor activity whose own per-event response is closer to the commercial-bucket value.

S8.4 Structural name-level checks

To move from spot-checks to panel-wide quantification, a verification exercise is conducted at the name-recurrence level. The 1.15M-POI 21-state catchment universe contains only 129,773 unique `location_name` values. The top 1,500 names cover 978,660 POIs. Each of the top names is classified as a national chain with consistent NAICS, a generic NAICS-category-as-name label, a public-sector entity, a brand-at-host pattern in which a service brand sits inside a host retailer, or a trade-association-at-members pattern in which an association name is spread across member-business addresses.

The verification identifies 62 names with bulk-recodable structural miscoding, of which 44 require an actual NAICS-3 change. The 44 effective recoding rules cover 63,924 POIs (5.5 percent of the 21-state universe by POI count and 5.3 percent of the panel by visitor-week count). The largest pattern is money-transfer-agent POIs that live inside grocery and convenience-store hosts. For the civic-versus-commercial decomposition, however, the implication is limited. Only 148 POIs would leave the civic bucket under the recoding rules, and the audit identifies no name that would move POIs from civic to commercial or commercial to civic. Re-estimating the category decomposition after applying the full set of structural recoding rules moves the high-baseline civic estimate from -0.135 to -0.120 , while the commercial estimate moves from -0.001 to $+0.001$. The order-of-magnitude civic-versus-commercial gap is preserved.

S8.5 Panel coverage

A separate concern is that businesses may be real but missing from the Advan POI universe altogether. If Advan systematically under-covers immigrant-serving businesses, the chilling regression would be attenuated downward in heavily Latino CBGs because the relevant POIs are not in the panel. This is tested using Veridion as an external firm registry. For each 5-digit ZIP in the panel, Advan POIs and Veridion firms are counted and the coverage ratio computed. Restricting to dense urban ZIPs and stratifying by foreign-born share quartile, the firm-weighted Advan/Veridion ratio is 0.28 in the lowest-FB quartile and 0.33 in the highest-FB quartile. The mild positive gradient does not support systematic undercoverage of immigrant neighborhoods.

Two Census establishment references corroborate the Veridion-based finding. At the county $\times$ NAICS-3 level, comparing Advan POIs to Census County Business Patterns, the establishment-weighted Advan/CBP coverage ratio rises monotonically from the lowest-FB quartile to the highest-

FB quartile. At the ZIP level, comparing against Census ZIP Business Patterns, the Advan/ZBP coverage ratio is essentially flat across foreign-born-share quartiles. Three independent external references therefore point the same direction. The panel-composition channel is a real concern in principle, but it is not the empirical source of the result.

S8.6 Residential labels and visit-pattern mismatches

A further exercise targets residential single-family homes that Advan labels with a generic NAICS-category string. A residence-mislabel flag is constructed, combining a generic NAICS-category-as-name label with a strict-residential street suffix. The flag identifies 28,204 POIs (2.4 percent of the 21-state panel). A stronger version that also requires no Veridion firm within 100 meters identifies 17,456 POIs (1.5 percent). Spot-checks confirm high precision. Only 157 of the flagged residences fall in the civic bucket, and dropping them would shrink the civic bucket by less than 1 percent. The remainder of the flag concentrates in the commercial and other buckets and does not shift the commercial estimate off its near-zero baseline.

A final check addresses visit-count series that are inconsistent with what the POI claims to be, using only the visit data itself. For each POI the Advan day-of-week and hour-of-day visit arrays are aggregated over all of 2025, each normalized to a shape profile, and a reference signature is built for every NAICS-3 with at least 30 well-measured POIs. A POI is flagged `flag_visit_pattern_mismatch` if both its day-of-week and hour-of-day profile correlate below 0.30 with its own sector's reference signature. The flag identifies 75,963 POIs (6.6 percent of the panel). It is the noisiest quality check. The flagged set mixes genuine anomalies with legitimate businesses that keep atypical hours. It is reported as a diagnostic rather than folded into the headline cleaning, because at this rate and with non-trivial false positives it is too blunt to use as a panel filter. Its value is that it catches a class of error that no external firm registry can detect, a real address with a real NAICS label but an implausible visit series.

References

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